ET AI HACKATHON 2026

MULTI-AGENT AUTONOMOUS SYSTEM

AgentFlow

Your AI Operations Team

8 AI agents that work together to complete real enterprise workflows — from employee onboarding to SLA breach prevention. When things break, they fix themselves. When it matters, they ask you first.

8	51+	100%	₹0
AI AGENTS	EDGE CASES	AUDITABLE	API COST

▶ agent-flow-amber.vercel.app

⚙ agentflow-api-788z.onrender.com

Team AttentionIsAllYouNeed

Bhaskar · Aditya · Harshit

LangGraph · Google Gemini AI · React · FastAPI · SQLite

Enterprise workflows are broken.

Three high-frequency workflows silently drain thousands of hours and lakhs of rupees every year in mid-size enterprises. They involve multiple teams, systems, and endless manual coordination.

Employee Onboarding

When a new employee joins, HR must manually create accounts across 4–6 systems (Workday, JIRA, Slack, Calendar, Email, ERP), assign a buddy, schedule orientations, and verify everything — involving 3–5 people across departments.

4–6 hours per hire × 200 hires/year = 1,000+ hours wasted

Meeting-to-Action Pipeline

After every team meeting, someone must manually extract action items, assign owners, create JIRA tasks, and send follow-ups. 40% of action items are forgotten within 48 hours (Harvard Business Review).

40% of action items lost · 45 min manual follow-up per meeting

SLA Breach Prevention

When a procurement approver goes on leave, approvals sit stuck for 24–48 hours. The SLA clock keeps ticking. Someone must find a delegate, reroute the approval, and document the override.

₹5,00,000/day in SLA penalties while approvals sit idle

Total annual cost of these broken workflows

₹83+ Lakhs

per year, for a 500-employee company

8 AI agents. One mission.

AgentFlow replaces manual coordination with specialized AI agents. Each has one job. Data flows through the pipeline. If something fails, the system heals itself. If it matters, it asks you first.

1

Signal Ingestor

Takes in the request. Extracts entities. Classifies urgency. Runs security scans against 20+ injection patterns.

2

Root Cause Analyzer

Queries enterprise systems to diagnose what's wrong. Detects cross-system conflicts, ghost entities, and data inconsistencies.

3

Strategy Planner

Creates multi-step execution plans with tool calls, owner assignments, and dependency ordering. Validates against hallucinated tools.

4

GPA Evaluator

Scores plan quality on 0.0–4.0 GPA scale. Detects contradictory reasoning. Plans below 2.0 are sent back for revision.

5

Human Review (HITL)

Pauses for human approval. You approve, reject, or ask questions. 5-min timeout → auto-escalation to supervisor.

6

Execution Engine

Calls HR, JIRA, Slack, Calendar, ERP. Self-heals on failure — exponential backoff and credential rotation.

7

Escalation Handler

Routes unrecoverable failures to the right human authority. Detects circular delegation loops and authority gaps.

8

Vernacular Reporter

Generates bilingual Hindi + English compliance report with edge case summary, audit trail, and executive overview.

Not a chatbot. An operations team.

Most AI tools answer questions. AgentFlow takes multi-step actions across real systems, handles failures, and maintains a complete audit trail.

🔧

Self-Healing Execution

When a tool fails — API timeout, 403, rate limit — the system retries with exponential backoff, credential rotation, and graceful degradation. The dashboard shows exact error codes and recovery actions.

👁️

Full Transparency

Every agent decision, tool call, and API response is logged live. A persistent SQLite database with 9 tables records every action for compliance review. No black box.

🧑

Human-in-the-Loop (HITL)

Before critical actions, the system pauses with a quality score. You approve, reject, or ask questions. 5-min timeout auto-escalates to supervisor — nothing stalls.

🛡️

51+ Edge Cases Handled

A centralized Edge Case Engine monitors 11 categories: duplicate identities, circular delegation, ghost entities, prompt injection, race conditions, and more.

💰

Cost-Efficient LLM Routing

Light Tier — Gemini Flash Lite

Used for: Signal Ingestor, Escalation Handler, Vernacular Reporter
Fast extraction tasks that don't need deep reasoning

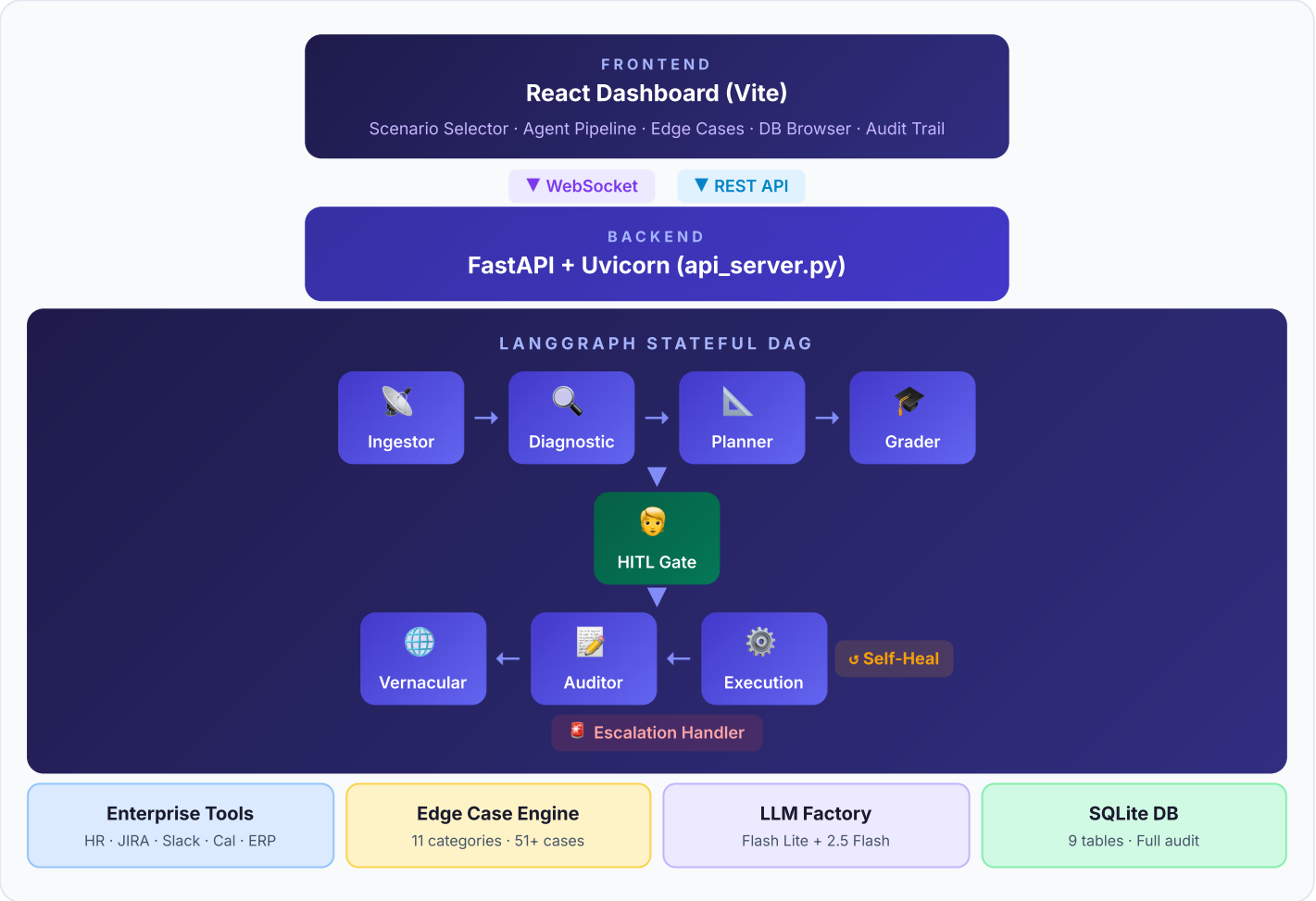
Heavy Tier — Gemini 2.5 Flash

Used for: Root Cause Analyzer, Strategy Planner, GPA Evaluator
Complex reasoning, planning, and quality assessment

Total API Cost: ₹0 (Google Gemini Free Tier)

How it's built.

A stateful directed agent graph orchestrated by LangGraph, with real-time WebSocket streaming to a React dashboard, backed by persistent SQLite storage.



LAYER	TECHNOLOGY	WHY THIS CHOICE
Orchestration	LangGraph (Stateful DAG)	State persistence, conditional routing, HITL interrupt
LLM	Google Gemini (Dual-Tier)	Free tier, fast inference, strong reasoning
Backend	FastAPI + WebSocket	Async, real-time streaming, auto API docs
Frontend	React (Vite)	Real-time pipeline visualization
Database	SQLite (9 tables)	Zero setup, embedded, portable
Deployment	Vercel + Render	Free tier, auto-deploy from GitHub

Quantified savings.

Back-of-envelope estimates for a mid-size enterprise (500–1,000 employees). Assumptions stated explicitly.

👤 Employee Onboarding

METRIC	BEFORE (MANUAL)	AFTER (AGENTFLOW)
Time per hire	4–6 hours	< 2 minutes
People involved	3–5 people	1 (approve only)
Error rate	~15%	< 1%

Annual Savings: ₹15,00,000 (200 hires × 5 hrs × ₹1,500/hr)

📋 Meeting-to-Action

METRIC	BEFORE (MANUAL)	AFTER (AGENTFLOW)
Follow-up time	45 minutes	< 1 minute
Action items lost	~40%	0% (auto-tracked)
JIRA creation	Manual, hours later	Instant, automatic

Annual Savings: ₹6,00,000 (600 meetings × 40 min × ₹1,500/hr)

⚡ SLA Breach Prevention

METRIC	BEFORE (MANUAL)	AFTER (AGENTFLOW)
Detection time	12–24 hours	Immediate
Resolution time	24–48 hours	< 2 minutes
Penalty exposure	₹5L/day	₹0

Penalties Avoided: ₹45,00,000 (12 events × 1.5 days × ₹5L/day × 50%)

TOTAL ANNUAL VALUE

₹83,10,000/year

Conservative estimate for a 500-employee enterprise

Operating Cost: ₹0/month (free tier infrastructure)

06 — EDGE CASE COVERAGE

51+ real-world scenarios handled.

A centralized Edge Case Engine monitors 11 categories during every workflow execution.

- **Identity & Entity**
  Duplicate names, ghost managers, cross-system inconsistency
- **Security Threats**
  Prompt injection (20+ patterns), SQL injection, data masking
- **Workflow Deadlock**
  Circuit breaker, planner loop, HITL timeout, plan overflow
- **Tool & API Failure**
  HTTP 403/429/504, auth expiry, credential rotation
- **Race Conditions**
  Duplicate workflow prevention, idempotency keys
- **LLM Guardrails**
  Hallucinated tools, contradictory reasoning, grade validation
- **Partial Failure**
  State consistency, dependency tracking, partial completion
- **HITL Resilience**
  5-min timeout, auto-escalation, WebSocket reconnection
- **Audit & Compliance**
  Full audit log, edge case log, tool call tracing
- **System & Infrastructure**
  Crash recovery, state persistence, LLM fallback
- **Scenario-Specific**
  Circular delegation, conflicting dates, ambiguous ownership

Ready for Evaluation

Pick any scenario, customize inputs, and watch 8 AI agents complete the workflow — with full transparency, self-healing, and human oversight.

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⌘ github.com/bhaskarkarn1/AgentFlow

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